



What is cloud computing

3 minutes

Cloud computing is the delivery of computing services over the internet. Computing services include common IT infrastructure such as virtual machines, storage, databases, and networking. Cloud services also expand the traditional IT offerings to include things like Internet of Things (IoT), machine learning (ML), and artificial intelligence (AI).

Because cloud computing uses the internet to deliver these services, it doesn't have to be constrained by physical infrastructure the same way that a traditional datacenter is. That means if you need to increase your IT infrastructure rapidly, you don't have to wait to build a new datacenter—you can use the cloud to rapidly expand your IT footprint.

Practical example

Suppose a retail team expects high traffic during a seasonal launch. Instead of buying and configuring extra physical servers months in advance, they can deploy additional compute capacity in the cloud for the launch period and scale down afterward. This approach improves agility and helps align spending to actual demand.

At a fundamentals level, cloud computing changes infrastructure planning from long procurement cycles to on-demand provisioning. Teams can test faster, recover faster, and adapt capacity as requirements change.

Cloud platforms also offer global reach, so teams can place services closer to users and design for regional resilience without building multiple physical datacenters.

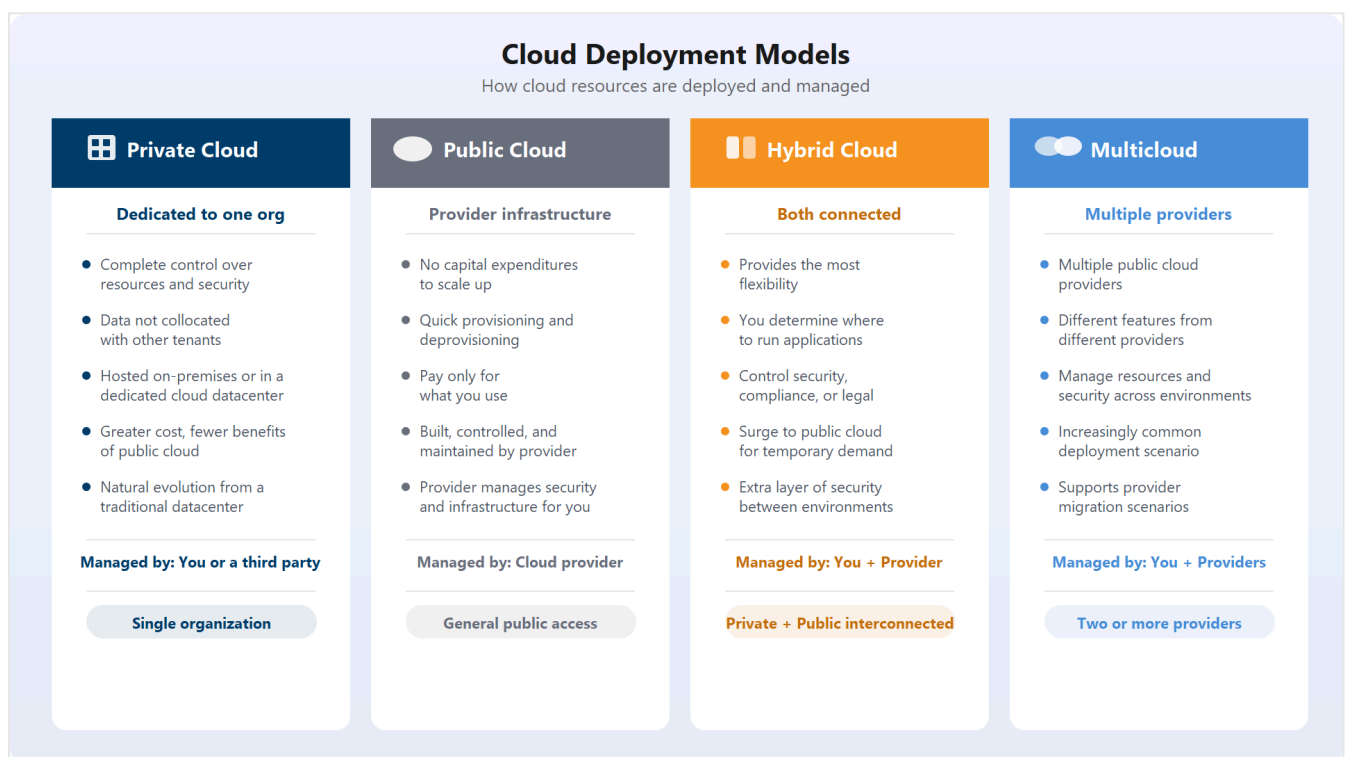
This short video provides a quick introduction to cloud computing.



Define cloud models

4 minutes

What are cloud models? The cloud models define the deployment type of cloud resources. The three main cloud models are: private, public, and hybrid.



Private cloud

A private cloud is a cloud environment used by a single entity. It evolved naturally from the traditional datacenter model, delivering IT services over the internet while keeping resources dedicated to one organization. Private cloud provides much greater control for your IT team. However, it also comes with greater cost and fewer of the benefits of a public cloud deployment. A private cloud may be hosted from your on-site datacenter, or in a dedicated datacenter offsite, potentially even by a third party.

Public cloud

A public cloud is built, controlled, and maintained by a third-party cloud provider. With a public cloud, anyone that wants to purchase cloud services can access and use resources. The general public availability is a key difference between public and private clouds.

Hybrid cloud

A hybrid cloud is a computing environment that uses both public and private clouds in an inter-connected environment. A hybrid cloud environment can be used to allow a private cloud to surge for increased, temporary demand by deploying public cloud resources. Hybrid cloud can be used to provide an extra layer of security. For example, users can flexibly choose which services to keep in public cloud and which to deploy to their private cloud infrastructure.

The following table highlights a few key comparative aspects between the cloud models.

 Expand table

Public cloud	Private cloud	Hybrid cloud
No capital expenditures to scale up	You have complete control over resources and security	Provides the most flexibility
Applications can be quickly provisioned and deprovisioned	Data isn't collocated with other tenants' data	You determine where to run your applications
You pay only for what you use	Hardware must be purchased for startup and maintenance	You control security, compliance, or legal requirements
You don't have complete control over resources and security	You are responsible for hardware maintenance and updates	

Multicloud

A fourth, and increasingly likely scenario is a multicloud scenario. In a multicloud scenario, you use multiple public cloud providers. Maybe you use different features from different cloud providers. Or maybe you started your cloud journey with one provider and are in the process

of migrating to a different provider. Regardless, in a multicloud environment you deal with two (or more) public cloud providers and manage resources and security in both environments.

Azure Arc

Azure Arc is a set of technologies that helps manage your cloud environment. Azure Arc can help manage your cloud environment whether it's a public cloud solely on Azure, a private cloud in your datacenter, a hybrid configuration, or even a multicloud environment running on multiple cloud providers at once.

Azure VMware Solution

What if you're already established with VMware in a private cloud environment but want to migrate to a public or hybrid cloud? Azure VMware Solution lets you run your VMware workloads in Azure with seamless integration and scalability.





Describe the shared responsibility model

3 minutes

You may have heard of the shared responsibility model, but you may not understand what it means or how it impacts cloud computing.

How responsibilities shift in cloud

Start with a traditional on-premises datacenter. Your team is responsible for maintaining the physical space, ensuring security, and maintaining or replacing the servers if anything happens. The IT department is responsible for maintaining all the infrastructure and software needed to keep the datacenter up and running. They're also likely to be responsible for keeping all systems patched and on the correct version.

With the shared responsibility model, these responsibilities get shared between the cloud provider and the consumer. Physical security, power, cooling, and network connectivity are the responsibility of the cloud provider. The consumer isn't collocated with the datacenter, so it wouldn't make sense for the consumer to have any of those responsibilities.

At the same time, the consumer is responsible for the data and information stored in the cloud. (You wouldn't want the cloud provider to be able to read your information.) The consumer is also responsible for access security, meaning you only give access to those who need it.

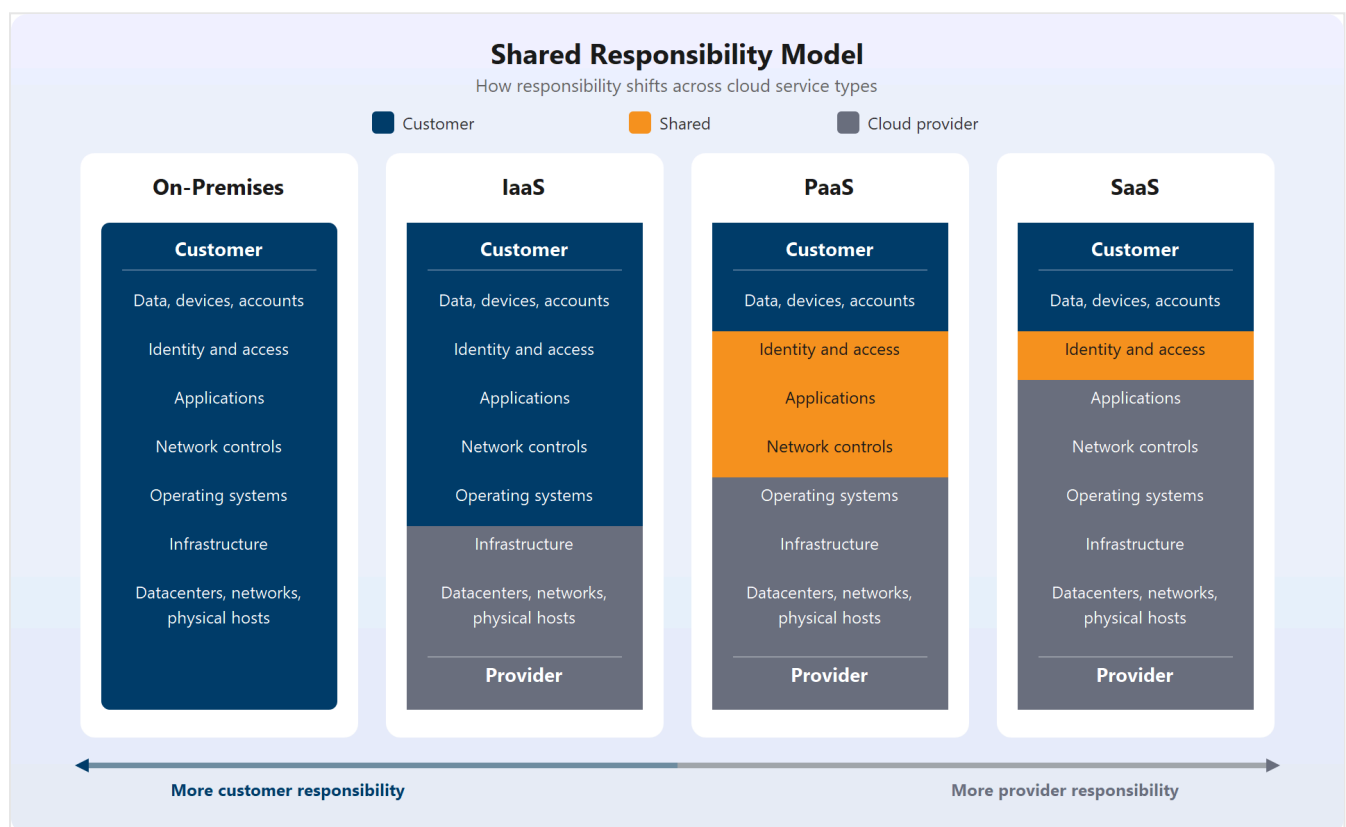
Then, for some things, the responsibility depends on the situation. If you're using a cloud SQL database, the cloud provider would be responsible for maintaining the actual database. However, you're still responsible for the data that gets ingested into the database. If you deployed a virtual machine and installed an SQL database on it, you'd be responsible for database patches and updates, as well as maintaining the data and information stored in the database.

With an on-premises datacenter, you're responsible for everything. With cloud computing, those responsibilities shift. The shared responsibility model is heavily tied into the cloud

service types (covered later in this learning path): infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS). IaaS places the most responsibility on the consumer, with the cloud provider being responsible for the basics of physical security, power, and connectivity. On the other end of the spectrum, SaaS places most of the responsibility with the cloud provider. PaaS, being a middle ground between IaaS and SaaS, rests somewhere in the middle and evenly distributes responsibility between the cloud provider and the consumer.

Responsibility by service model

The following diagram highlights how the Shared Responsibility Model informs who is responsible for what, depending on the cloud service type.



When using a cloud provider, you'll always be responsible for:

What always stays with you

- The information and data stored in the cloud
- Devices that are allowed to connect to your cloud (cell phones, computers, and so on)
- The accounts and identities of the people, services, and devices within your environment

The cloud provider is always responsible for:

What the provider always owns

- The physical datacenter
- The physical network
- The physical hosts

Your service model will determine responsibility for things like:

What depends on the service type

- Operating systems
- Network controls
- Applications
- Identity and access
- Infrastructure

For example, identity and access is shared in PaaS and SaaS—you manage your own users, roles, and policies, while the provider runs the authentication platform (such as Microsoft Entra ID). Infrastructure, on the other hand, shifts entirely to the provider as soon as you move off-premises to IaaS.



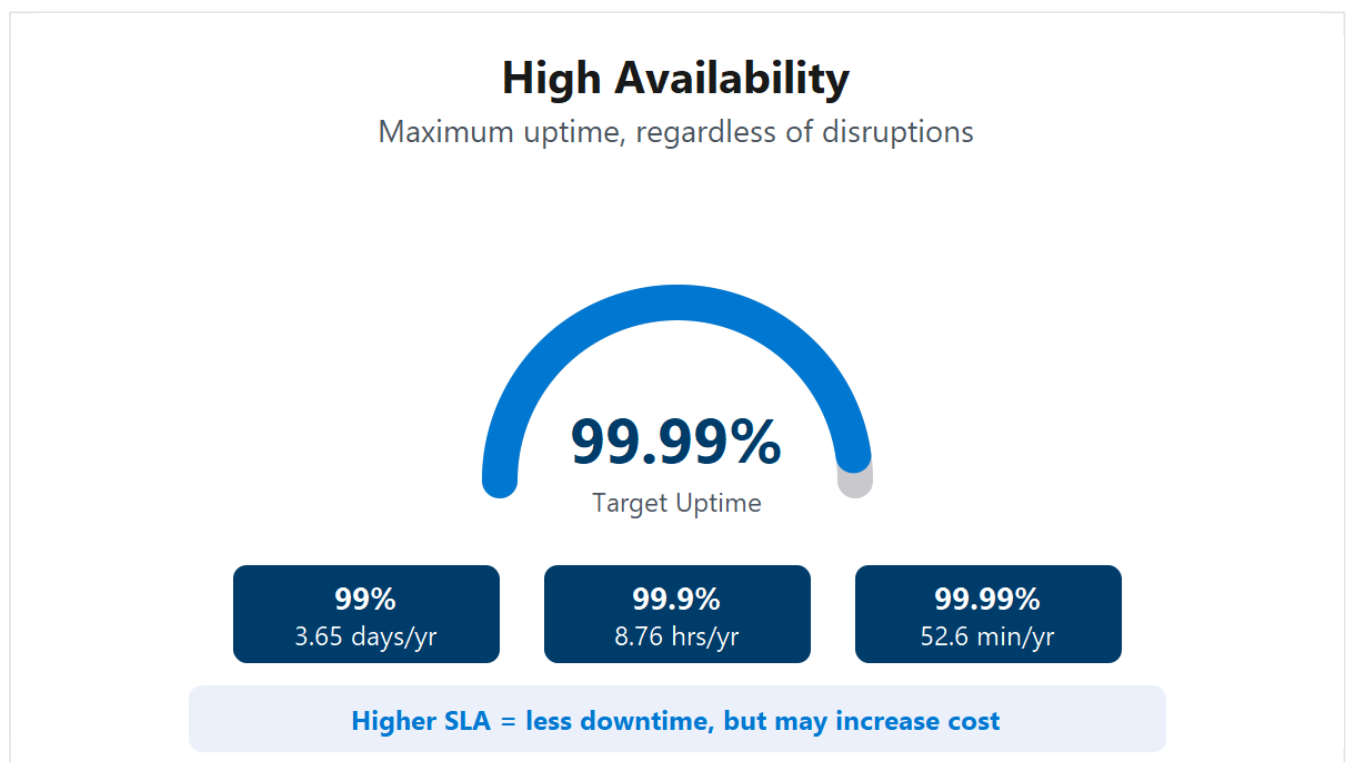


Describe the benefits of high availability and scalability in the cloud

5 minutes

When building or deploying a cloud application, two of the biggest considerations are uptime (or availability) and the ability to handle demand (or scale).

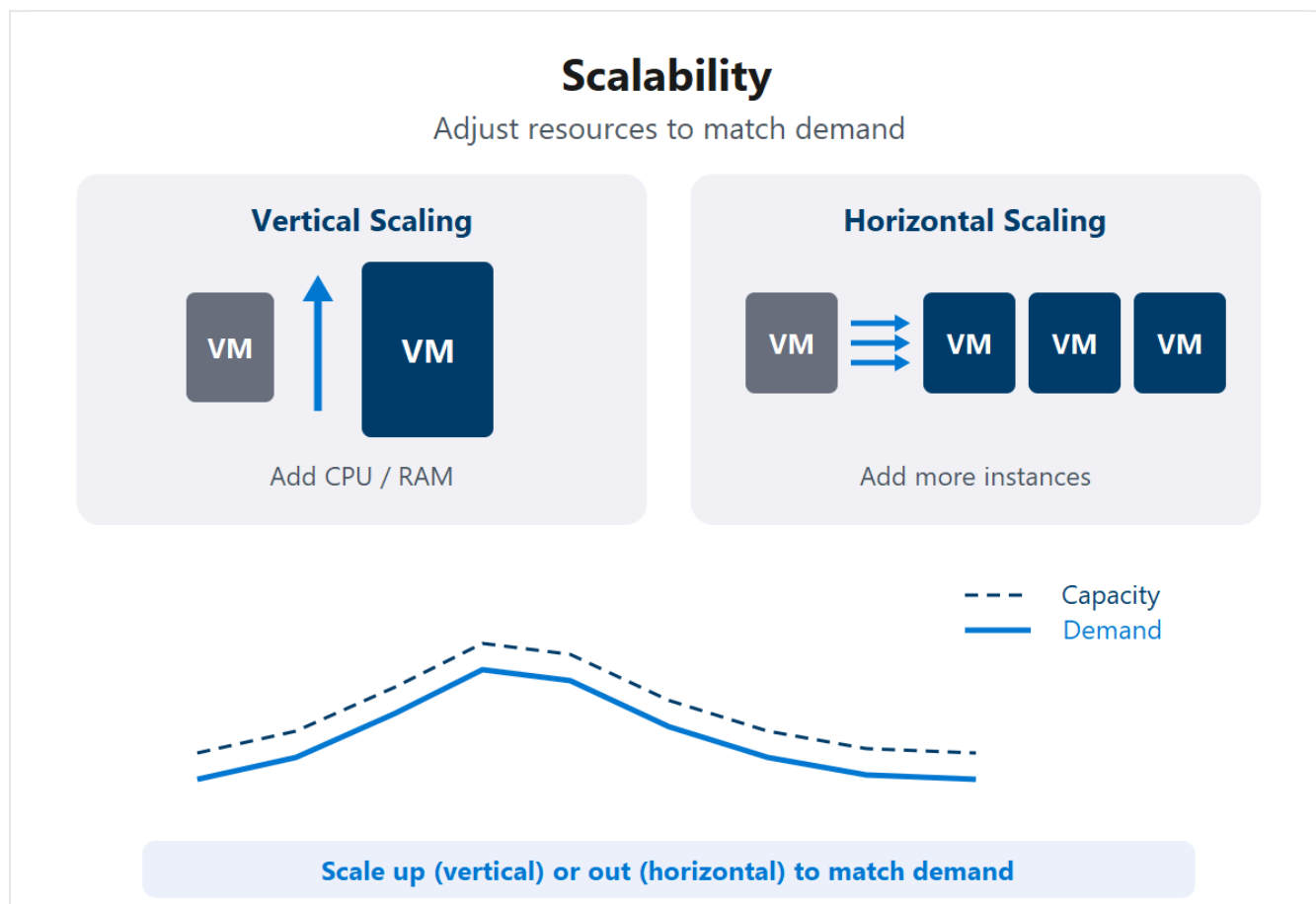
High availability



When you're deploying an application, a service, or any IT resources, it's important the resources are available when needed. High availability focuses on ensuring maximum availability, regardless of disruptions or events that may occur.

When you're architecting your solution, you need to account for service availability guarantees. Azure is a highly available cloud environment with uptime guarantees depending on the service. These guarantees are part of the service-level agreements (SLAs).

Scalability



Another major benefit of cloud computing is the scalability of cloud resources. Scalability refers to the ability to adjust resources to meet demand. If you suddenly experience peak traffic and your systems are overwhelmed, the ability to scale means you can add more resources to better handle the increased demand.

The other benefit of scalability is that you aren't overpaying for services. Because the cloud is a consumption-based model, you only pay for what you use. If demand drops off, you can reduce your resources and thereby reduce your costs.

Scaling generally comes in two varieties: vertical and horizontal. Vertical scaling is focused on increasing or decreasing the capabilities of resources. Horizontal scaling is adding or subtracting the number of resources.

Vertical scaling

With vertical scaling, if you were developing an app and you needed more processing power, you could vertically scale up to add more CPUs or RAM to the virtual machine. Conversely, if

you realized you had over-specified the needs, you could vertically scale down by lowering the CPU or RAM specifications.

Horizontal scaling

With horizontal scaling, if you suddenly experienced a steep jump in demand, your deployed resources could be scaled out (either automatically or manually). For example, you could add additional virtual machines or containers. In the same manner, if there was a significant drop in demand, deployed resources could be scaled in (either automatically or manually).

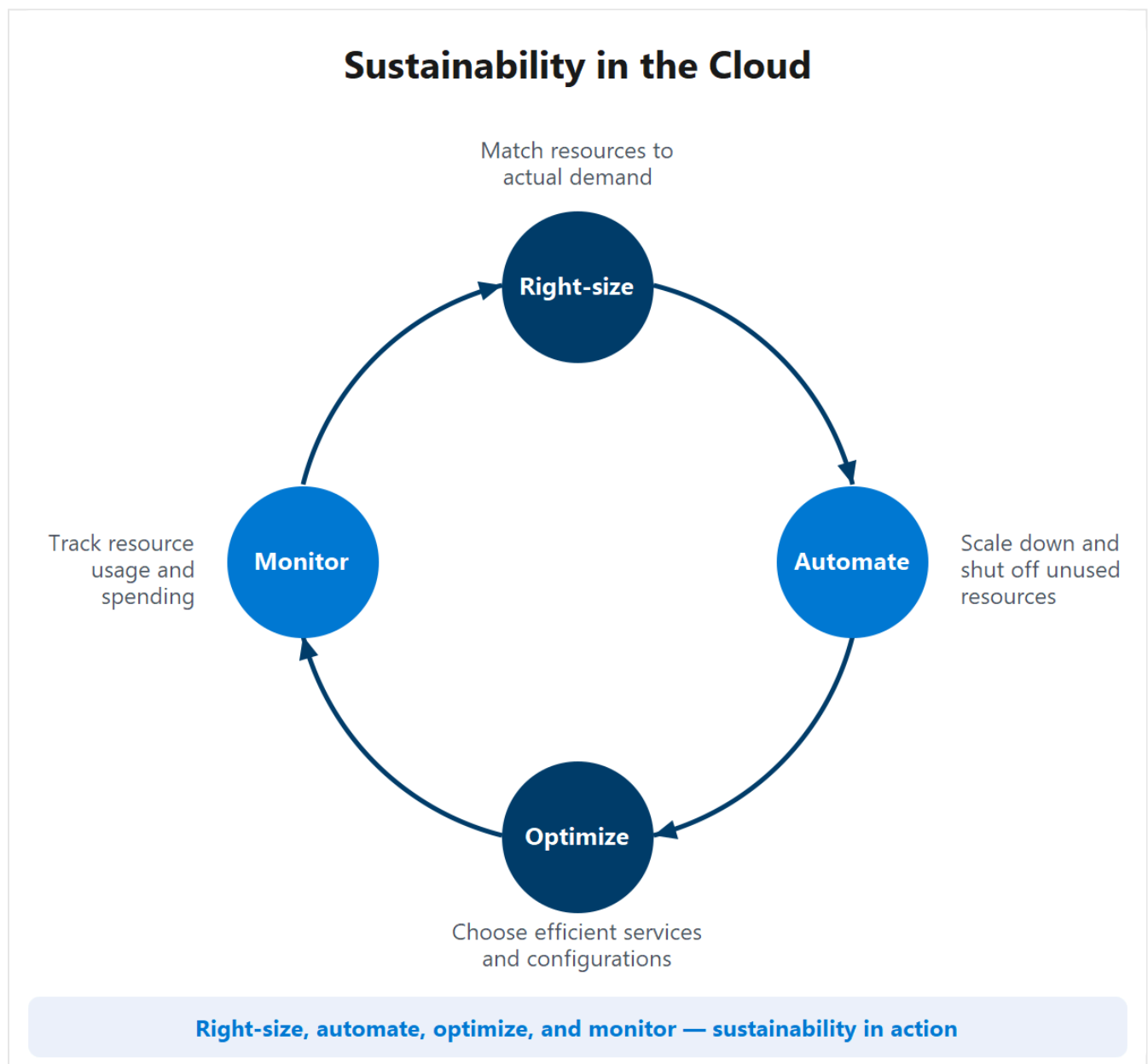




Describe sustainability considerations in the cloud

2 minutes

Cloud computing can support sustainability goals when teams actively optimize how resources are deployed and used.



Why cloud can improve efficiency

Cloud providers operate at large scale, which can improve resource utilization compared to many isolated on-premises environments. In Azure, you can also reduce waste by matching deployed resources to actual demand.

Sustainability-aligned cloud practices

Examples of practices that support sustainability and cost efficiency include:

- Scaling resources down when demand decreases
- Turning off or deallocating resources that are not in use
- Choosing efficient services and configurations to reduce overprovisioning
- Using governance and monitoring to track usage trends and optimize deployments over time

For example, a development environment that runs only during business hours can be automatically shut down overnight and on weekends. This practice reduces unnecessary consumption while still meeting team needs.

At a fundamentals level, sustainability in the cloud is closely tied to good operational habits: right-size, automate, monitor, and continuously optimize.

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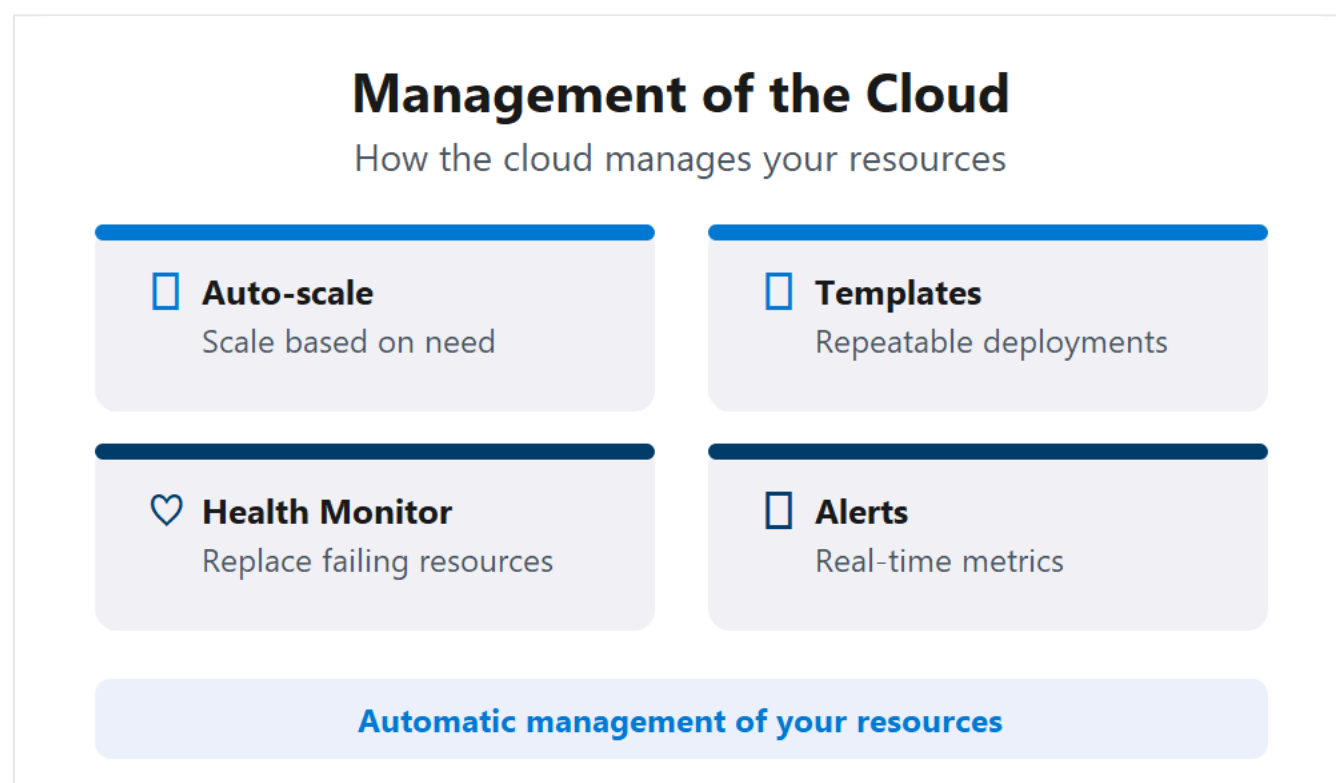


Describe the benefits of manageability in the cloud

2 minutes

A major benefit of cloud computing is the manageability options. There are two types of manageability for cloud computing, and both are excellent benefits.

Management of the cloud

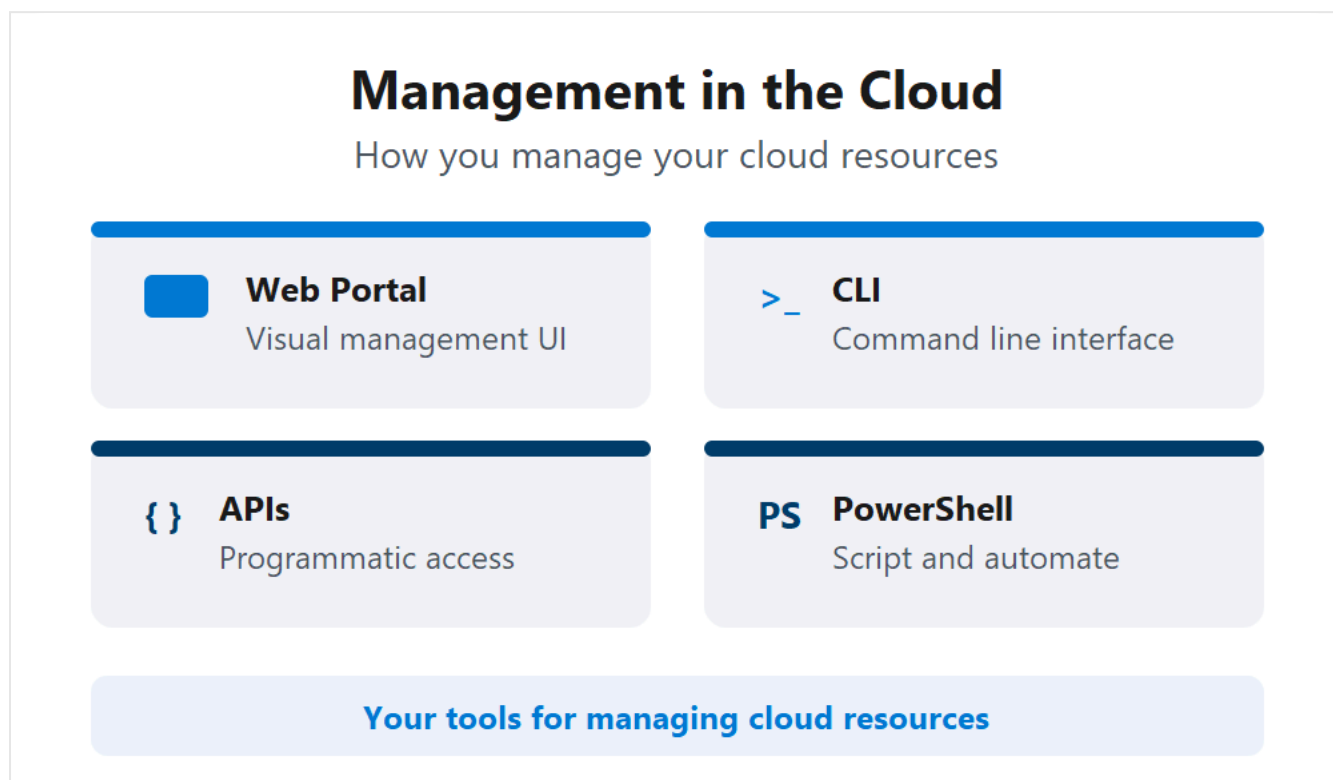


Management of the cloud speaks to managing your cloud resources. In the cloud, you can:

- Automatically scale resource deployment based on need.
- Deploy resources based on a preconfigured template, removing the need for manual configuration.
- Monitor the health of resources and automatically replace failing resources.

- Receive automatic alerts based on configured metrics, so you're aware of performance in real time.

Management in the cloud



Management in the cloud speaks to how you're able to manage your cloud environment and resources. You can manage these:

- Through a web portal.
- Using a command line interface.
- Using APIs.
- Using PowerShell.

For example, an operations team can deploy resources from templates, monitor health in the portal, and automate recurring tasks with CLI or PowerShell scripts. This combination reduces manual effort and helps maintain consistent configurations.

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Describe the consumption-based model

3 minutes

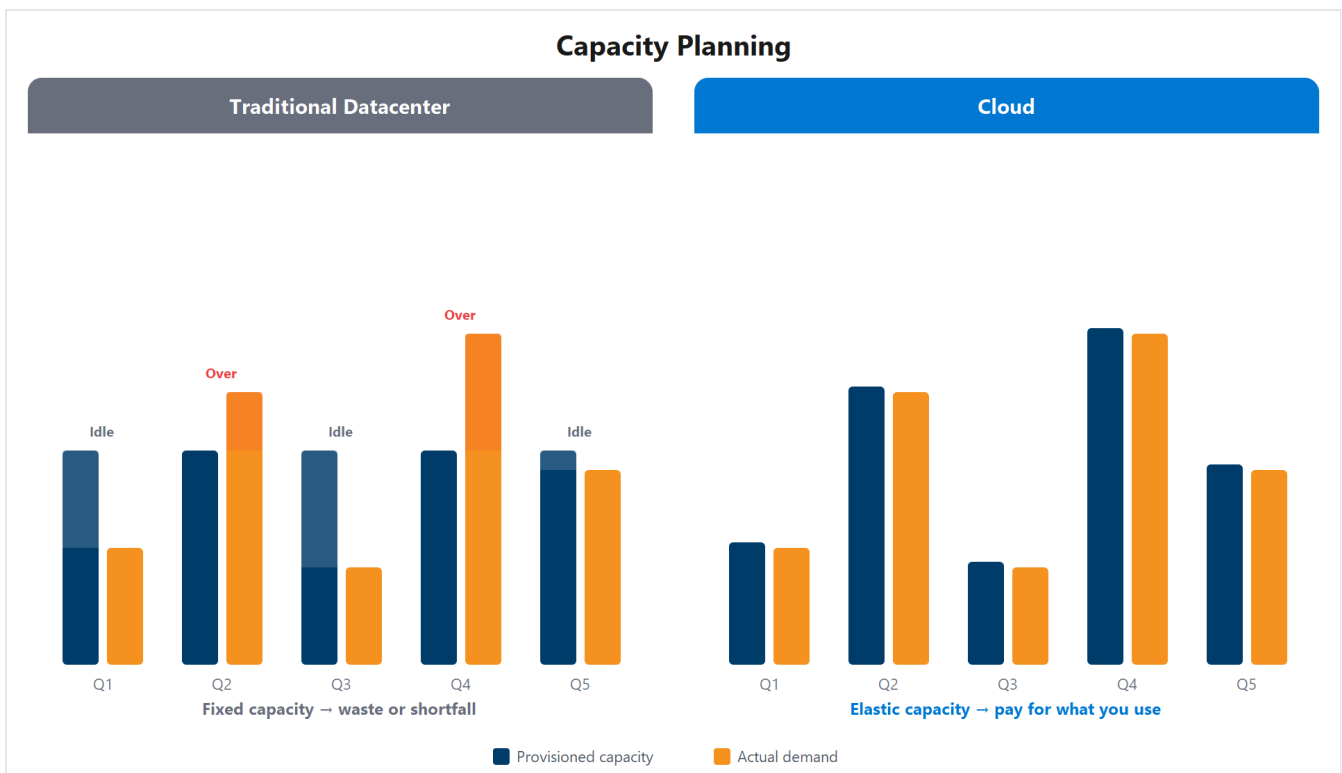
Cloud computing operates on a consumption-based model. You pay for the IT resources you use, and nothing more. Instead of buying and maintaining your own datacenter infrastructure, you rent compute power and storage and release those resources when you're done.

In traditional IT budgeting, you may hear the terms capital expenditure (CapEx) and operational expenditure (OpEx). CapEx refers to up-front spending on physical infrastructure like servers, network hardware, and datacenter space. OpEx refers to ongoing spending on services over time. Because you pay for cloud services as you consume them, cloud computing is classified as an operational expense.

The consumption-based model offers several key benefits:

- No upfront costs for hardware or datacenter infrastructure.
- No need to purchase and manage capacity that may go underutilized.
- The ability to add resources when demand increases.
- The ability to release resources when demand decreases.

Capacity planning: traditional vs. cloud



With a traditional datacenter, you estimate future resource needs in advance. Overestimate, and you overspend on infrastructure that sits idle. Underestimate, and your applications suffer degraded performance. Fixing the problem means ordering, installing, and provisioning additional hardware, power, and cooling.

In a cloud-based model, you adjust resources to match actual demand. Add virtual machines when you need more capacity; remove them when demand drops. You pay only for what you use, not for idle capacity. In practice, you can scale out during peak demand and scale back in when traffic subsides.

Cloud pricing

Cloud providers use a pay-as-you-go pricing model. You typically pay only for the services you consume, which helps you:

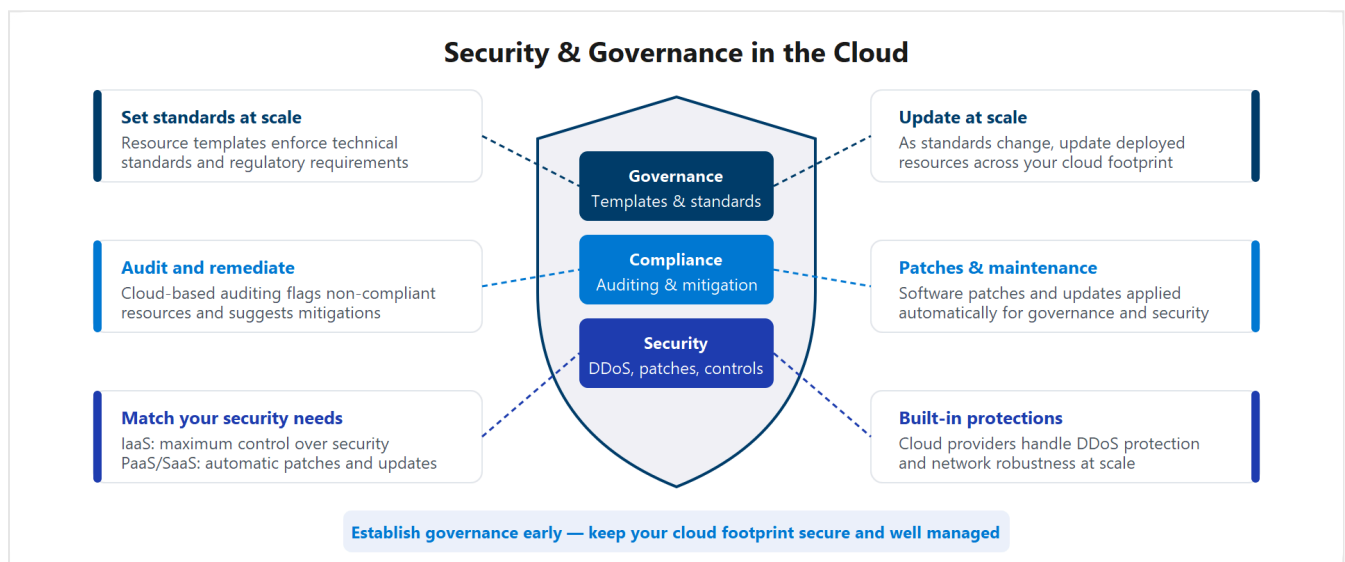
- Plan and manage operating costs.
- Run infrastructure more efficiently.
- Scale as workload needs change.

The cloud provider maintains the underlying infrastructure, including power, cooling, hardware, and networking, so you can focus on solving business problems and delivering new capabilities to your users.

Describe the benefits of security and governance in the cloud

2 minutes

Whether you're deploying infrastructure as a service or software as a service, cloud features support governance and compliance.



Tools such as templates help ensure that deployed resources meet your technical standards and regulatory requirements. As standards change, you can update resources at scale. Cloud-based auditing helps flag resources that are out of compliance with your baseline and provides mitigation strategies. Depending on your operating model, software patches and updates may also be applied automatically, which helps with both governance and security.

On the security side, you can find a cloud solution that matches your security needs. If you want maximum control of security, infrastructure as a service provides you with physical resources but lets you manage the operating systems and installed software, including patches and maintenance. If you want patches and maintenance taken care of automatically, platform as a service or software as a service deployments may be the best cloud strategies for you.

Cloud providers are typically well suited to handle things like distributed denial of service (DDoS) attacks, making your network more robust and secure.

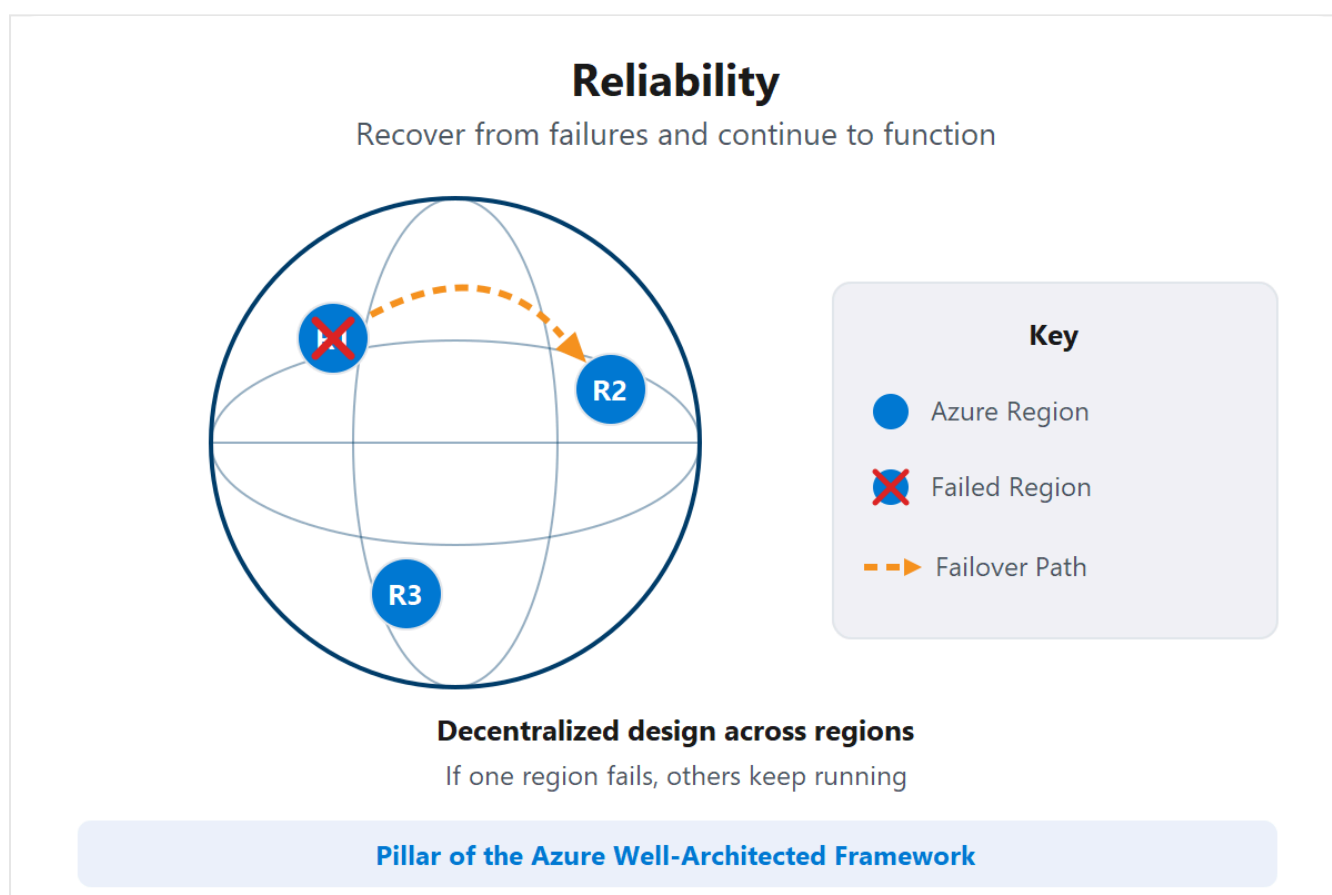


Describe the benefits of reliability and predictability in the cloud

2 minutes

Reliability and predictability are two crucial cloud benefits that help you develop solutions with confidence.

Reliability

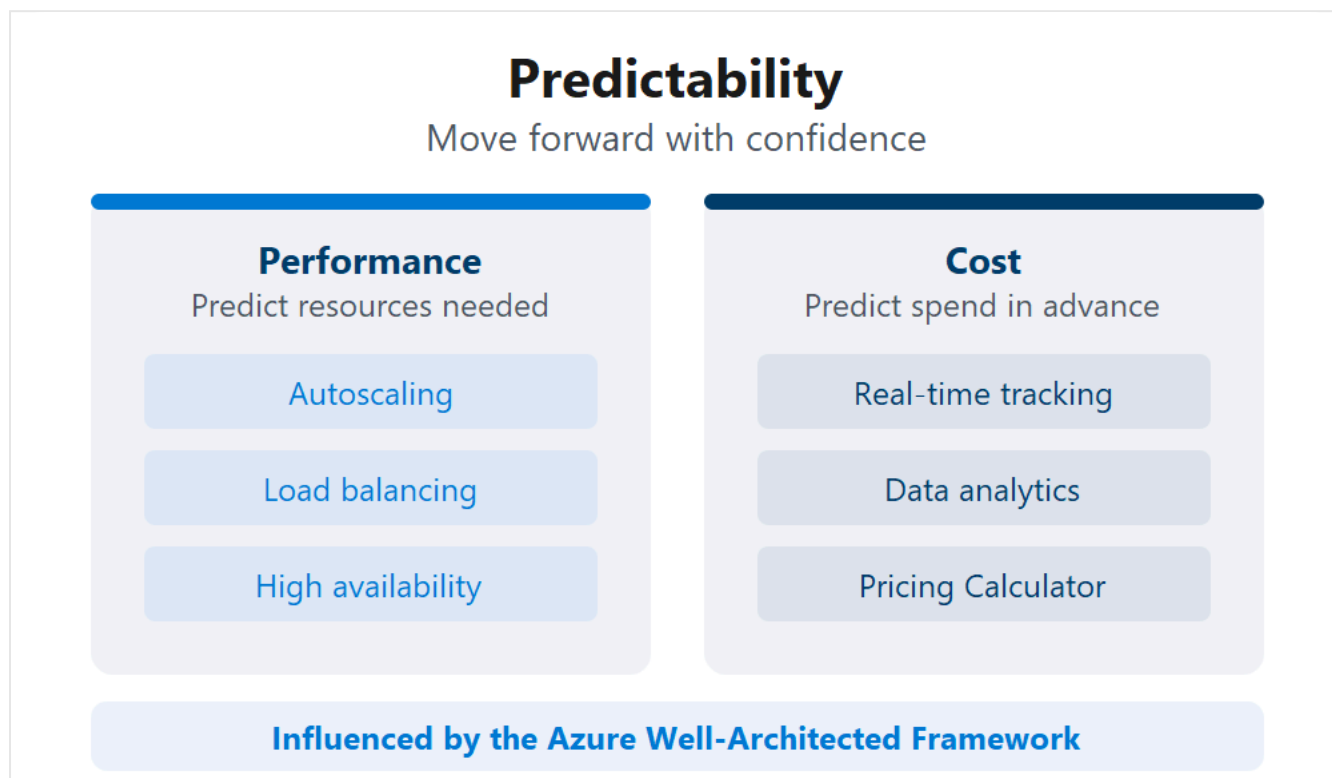


Reliability is the ability of a system to recover from failures and continue to function. It's also one of the pillars of the Microsoft Azure Well-Architected Framework.

The cloud, by virtue of its decentralized design, naturally supports a reliable and resilient infrastructure. With a decentralized design, the cloud enables you to have resources deployed

in regions around the world. With this global scale, even if one region has a catastrophic event other regions are still up and running. You can design your applications to automatically take advantage of this increased reliability. In some cases, your cloud environment itself will automatically shift to a different region for you, with no action needed on your part. You'll learn more about how Azure leverages global scale to provide reliability later in this series.

Predictability



Predictability in the cloud lets you move forward with confidence. Predictability can be focused on performance predictability or cost predictability. Both performance and cost predictability are heavily influenced by the Microsoft Azure Well-Architected Framework. Deploy a solution built around this framework and you have a solution whose cost and performance are predictable.

Performance

Performance predictability focuses on predicting the resources needed to deliver a positive experience for your customers. Autoscaling, load balancing, and high availability are just some of the cloud concepts that support performance predictability. If you suddenly need more resources, autoscaling can deploy additional resources to meet the demand, and then scale back when the demand drops. Or if the traffic is heavily focused on one area, load balancing will help redirect some of the overload to less stressed areas.

Cost

Cost predictability is focused on predicting or forecasting the cost of the cloud spend. With the cloud, you can track your resource use in real time, monitor resources to ensure that you're using them in the most efficient way, and apply data analytics to find patterns and trends that help better plan resource deployments. By operating in the cloud and using cloud analytics and information, you can predict future costs and adjust your resources as needed. You can even use tools like the Azure Pricing Calculator to get an estimate of potential cloud spend.

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Summary

2 minutes

In this module, you learned about some of the benefits of operating in the cloud. You learned about high availability and reliability, and how those work to keep your applications running. You also learned about how the cloud can provide a more secure environment and support sustainability goals. Finally, you learned that the cloud provides a highly manageable environment for your resources.

Learning objectives

You should now be able to:

- Describe the benefits of high availability and scalability in the cloud.
- Describe the benefits of reliability and predictability in the cloud.
- Describe the benefits of security and governance in the cloud.
- Describe the benefits of manageability in the cloud.
- Describe sustainability considerations and benefits in the cloud.

Additional resources

The following resources provide more information on topics in this module or related to this module.

- [Build great solutions with the Microsoft Azure Well-Architected Framework](#) - A Microsoft Learn course that introduces you to the Microsoft Azure Well-Architected Framework.

Explore with Copilot



Tip

Try one of these prompts in Copilot Chat:

- "Evaluate an on-premises system and map cloud improvements for availability, scalability, reliability, security, and manageability."
- "Design a KPI dashboard to measure whether cloud adoption is delivering the expected technical and business benefits."
- "Analyze tradeoffs where cloud benefits may be limited, including cost, governance complexity, and sustainability constraints."

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Summary

2 minutes

In this module, you learned about general cloud concepts. You started with an understanding of what cloud computing is. You also learned about the shared responsibility model and how you and your cloud provider share the responsibility of keeping your information in the cloud secure. You covered the differences between the cloud models (public, private, hybrid, and multicloud). Then, you wrapped up with a unit on how the cloud shifts IT spend from a capital expense to an operational expense.

Learning objectives

You should now be able to:

- Define cloud computing.
- Describe the shared responsibility model.
- Define cloud models, including public, private, and hybrid.
- Identify appropriate use cases for each cloud model.
- Describe the consumption-based model.
- Compare cloud pricing models.

Additional resources

The following resources provide more information on topics in this module or related to this module.

- [Shared responsibility model](#) - The sharing of responsibilities for the cloud between you and your cloud provider.
- [Introduction to Azure VMware Solution](#) - A Microsoft Learn course that dives deeper into Azure VMware Solution.
- [Introduction to Azure hybrid cloud services](#) - A Microsoft Learn course that explains hybrid cloud in greater detail.

Explore with Copilot

Tip

Try one of these prompts in Copilot Chat:

- "Explain shared responsibility across on-premises, IaaS, PaaS, and SaaS using one concrete business application example."
- "Compare public, private, hybrid, and multicloud models, then recommend the best fit for three different organizational scenarios."
- "Create a financial comparison of CapEx versus OpEx for moving a workload to a consumption-based cloud model."

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